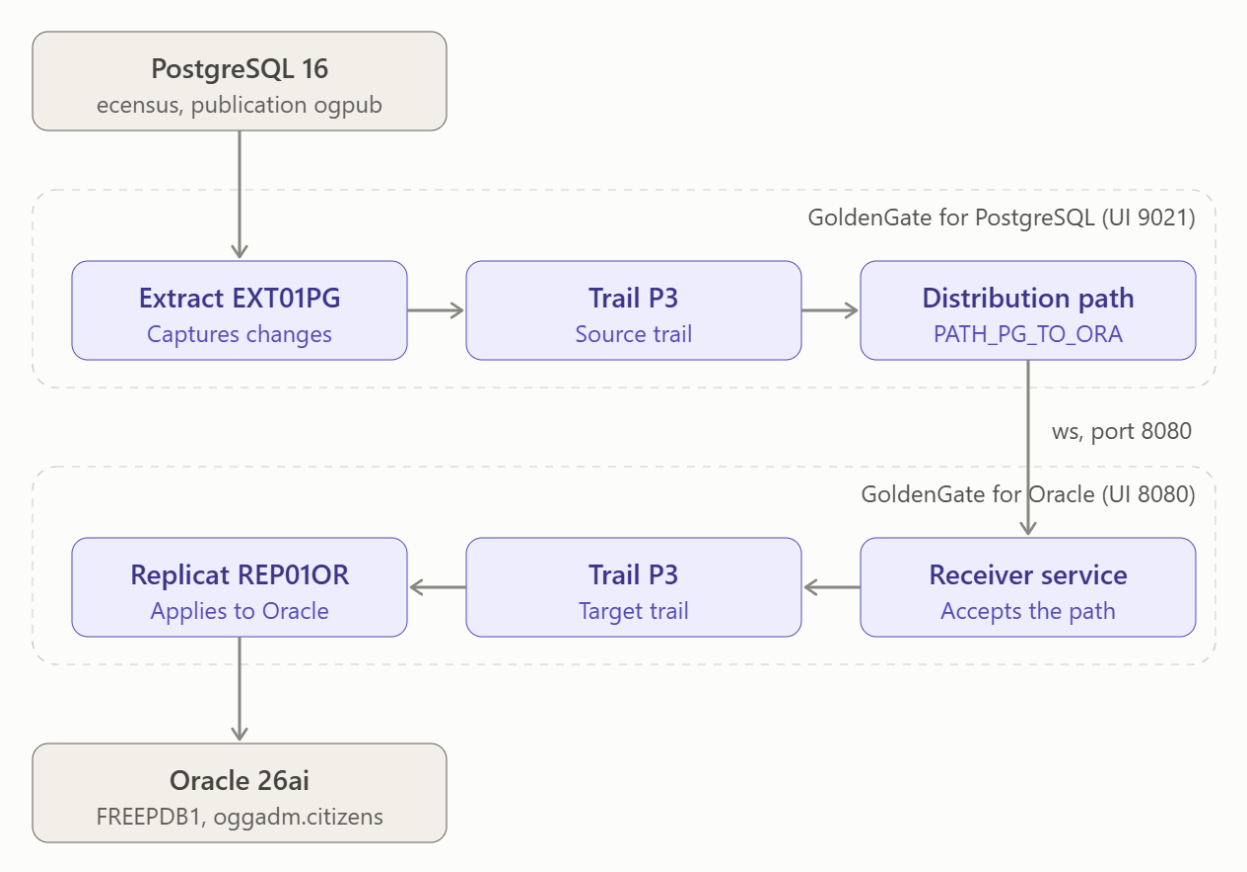


PostgreSQL to Oracle replication with Oracle GoldenGate

Setup notes, lessons learned and LinkedIn draft. Oracle GoldenGate 23ai for PostgreSQL and for Oracle, running in Docker.

Architecture overview

Changes flow in one direction through these hops:

1. PostgreSQL 16 (database ecensus, publication ogpub)
2. GoldenGate for PostgreSQL (UI port 9021): Extract EXT01PG, then source trail P3, then distribution path PATH_PG_TO_ORA
3. Websocket (ws) over port 8080 to the target
4. GoldenGate for Oracle (UI port 8080): Receiver service, then target trail P3, then Replicat REP01OR
5. Oracle 26ai Free, PDB FREEPDB1, table oggadm.citizens

Main points to keep in mind

1. **Trail names must match at every hop.** The Extract writes trail P3. The path reads source trail P3 and writes target trail P3. The Replicat reads P3. If any one differs, every process still shows running while no data moves.
2. **Running does not mean data is flowing.** Check the statistics and the trail file sizes on both sides. A trail of about 1.4 KB holds only the header.
3. **The path needs its own credential.** The distribution path uses a GoldenGate login for the target deployment, stored in Path Connections (domain Network) on the source. The Oracle database login belongs to the Replicat and is a separate credential.
4. **GoldenGate captures only changes made after the Extract starts.** Existing rows, and rows inserted while the Extract was down, need a separate load. Start the Extract first, then load the existing data.
5. **Never delete trail files while a checkpoint still points at them.** This caused the corrupted-trail abend and the later "no trail file found" failure. Delete the process and its trail together, or leave them alone.
6. **Use an address that works from inside the container.** Inside a container, localhost refers to the container itself. Use the host gateway or the target container IP, and test it with curl first.
7. **Start and stop in the right order.** Start: Extract, then path, then Replicat. Stop: path, Replicat, then Extract. Avoid restarting containers while data is being written.
8. **Do not copy EXTRACT or REPLICAT lines into the parameter file.** The UI adds them, and the duplicates only produce warnings.

Recommended configuration

Item	Value
PostgreSQL	wal_level=logical, a publication (ogpub) on the table, a primary key on the table
Extract	USERIDALIAS, TRANLOGOPTIONS STREAMINGOPTIONS (publication_names="ogpub";), EXTTRAIL P3, TABLE public.citizens;
Path	Source EXT01PG trail P3, target ws://<target-ip>:8080, target trail P3, alias in the Network domain

Replicat	Integrated, trail P3, ASSUMETARGETDEFS, MAP public.citizens, TARGET OGGADM.CITIZENS;
After the initial load	Remove HANDLECOLLISIONS and restart the Replicat

High-level steps

1. Prepare PostgreSQL: set wal_level=logical, create the publication, and create the GoldenGate user.
2. Prepare Oracle: create the target schema and table, and the GoldenGate user in the PDB.
3. Deploy both GoldenGate containers and create the database credentials in each.
4. Create the Extract on the source and start it.
5. Do the initial load of the existing rows into Oracle.
6. Create the Path Connection alias with the target GoldenGate login.
7. Create the distribution path and start it.
8. Create the Replicat with the matching trail name and start it.
9. Verify insert, update and delete, then remove HANDLECOLLISIONS.

Verification checklist

- Extract statistics show inserts on public.citizens.
- Source trail file grows after a new insert.
- Path statistics show LCR sent increasing.
- Target trail file grows.
- Replicat statistics show inserts with no conflicts or discards.
- Row counts match on both databases.